

Project Workflow: Week 5

Phase: Paddle Movement & Controls

1. Objective

The primary goal of Week 5 was to implement smooth and responsive paddle movement for the Ping Pong game. This included:

- Implementing Player 1 and Player 2 paddle controls.
- Using keyboard input for real-time paddle movement.
- Ensuring paddles move smoothly without leaving the game area.
- Connecting the paddle controls with the existing game interface.

2. Detailed Task Breakdown

- Player 1 Controls: Implemented W and S keyboard controls to move the first player's paddle upward and downward.
- Player 2 Controls: Implemented Up Arrow and Down Arrow keyboard controls for the second player's paddle.
- Keyboard Input Handling: Used Java KeyListener to detect keyboard input and provide responsive paddle movement.
- Boundary Control: Added movement restrictions so that the paddles remain inside the game play area.
- Movement Testing: Tested paddle movement at different positions and ensured that the controls respond correctly without unwanted movement.

3. Deliverables

- Functional Player 1 paddle controls.
- Functional Player 2 paddle controls.
- Keyboard input handling system.
- Paddle boundary restriction.
- Tested and working paddle movement.

4. Tools & Resources

- Programming Language: Java
- GUI Framework: Java Swing
- Input Handling: KeyListener
- Development Environment: IntelliJ IDEA / VS Code
- Reference Materials: Java KeyListener and Swing documentation